

Carve Engineering Spec

A *living* document describing the **Carve** pipeline. See the [spec](#) and `Carve.parse` for the AST.

Architecture

The parser runs in **two passes**:

1. Block identification (whole document)
2. Inline parsing (delimiter stack, no backtracking)
 - links and images
 - emphasis: **strong**, *emphasis*, ***bold italic***
 - `inline code`

Design principle: plain text should render beautifully and stay readable at the source.

Carve design doc

Engines

Engine	Language	Status
carve-js	TypeScript	reference
carve-rs	Rust	reference
carve-php	PHP	full
carve-rb	Ruby (FFI)	full

Task status

- [x] Block parser
- [x] Inline parser
- [x] HTML renderer
- [] PDF renderer polish
 - [x] tables
 - [x] math images

Notes

Term : A single unit of markup meaning.

AST : The parsed tree returned by `Carve.parse`.

```
pub fn parse(source: &str) -> Document {  
  // block pass, then inline pass  
  document::from(source)  
}
```

Built with the pure-Ruby HexaPDF engine.